

Instructor: Date:

Learner: Regional Branch:

Program: Center:

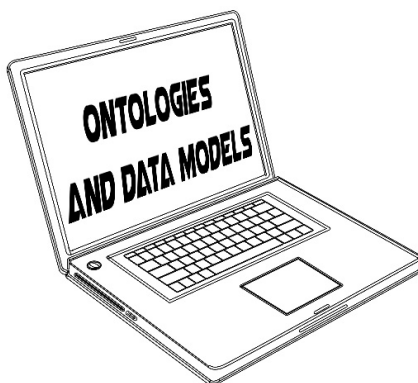
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Let's Start!

Focusing on Data and Meaning



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Read and write about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can read about ontologies and data models.
- I can write about ontologies and data models to explain what they are.
- I can understand what ontologies and data models are.

Ask your partner:

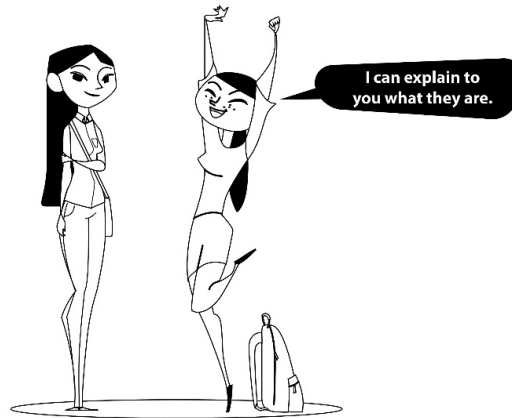
- How does a computer program recognize the difference between two objects or classes?
- What is an effective method for establishing classes and subclasses when writing code?





Let's Explore!

Sonia explains what the difference between ontologies and data models is.



Instructions: Read the text below.

- Clara:** Hi, Sonia. Did you study for the Artificial Intelligence test?
- Sonia:** Yes, I did. I didn't have to work, so I could study the whole weekend; as a result, I think I am ready. Do you have any questions?
- Clara:** Well, I can't understand what ontologies and data models are. In other words, I think they are the same.
- Sonia:** I can see why you think that; people confuse those concepts easily.
- Clara:** Can you please explain to me the difference between an ontology and a data model?
- Sonia:** Sure! An ontology is a view of the concepts, relations, and rules for a particular area of business information. This is an abstract view that doesn't take into account how that information can be stored as data; therefore, an ontology consists of classes to which relevant objects belong, regardless of storage location.
- Clara:** And what about data models? Can you tell me what they are?
- Sonia:** On the other hand, we can create data models to represent the business reality, providing a design for data storage location and its management.
- Clara:** Now I see why I couldn't differentiate both concepts. Can you give me an example of an ontology?
- Sonia:** Sure. For instance, a business ontology can include the following classes: ENTITY, defined as a business or person; SUPPLIER, defined as an ENTITY that provides goods or services; Finally, CUSTOMER, defined as an ENTITY that buys the company's goods or services. As an example, we can see that the last two classes include as their parent or "superclass," the ENTITY class. In other words, we can say the ENTITY class includes the last two classes.
- Clara:** Thanks for your help. Now I think I will be able to pass the exam tomorrow.



Let's Practice!

Are ontologies and data models the same?

Instructions: Read the paragraph. Underline all the mistakes in the paragraph.

Some people don't know the difference between ontology and data models, there it is important for you to know it. Ontology helps to get an abstract view of concepts, rules, and relations. In the other hand, data models represent a business reality, providing a design for data storage location and its management. On others words, both are related but they're not the same.



Let's Practice!

It's important to understand these concepts.

Instructions: Match column A with column B.

COLUMN A

1. People can confuse
2. We can represent
3. Thanks to ontology

COLUMN B

- a. a business reality with data models.
- b. these two concepts easily.
- c. we can view concepts, relations, and rules.





Extension Activity

A Synoptic Table

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Look up more information about data models and ontology.
- Design a synoptic table in which you explain the concepts of data models and ontology.
- Share your table with the class.

Self-Assessment

Yes

No

- I read about data models and ontology.
- I differentiate data models from ontology.
- I design a synoptic table to explain data models and ontology.
- I share my work with the class.

☐	☐
☐	☐
☐	☐
☐	☐



Target Words

- **concept** (noun): idea or notion
e.g. Now I see why I couldn't differentiate both concepts.
- **belong** (verb): to be a part of something
e.g. An ontology consists of classes to which relevant objects belong, regardless of storage location.
- **superclass** (noun): superior category that includes smaller categories or classes
e.g. We can see that the last two classes include as their parent or "superclass," the ENTITY class.
- **same** (adjective): equal or identical
e.g. In other words, I think they are the same.
- **easily** (adverb): in an easy way, without complications
e.g. People confuse those concepts easily.
- **entity** (noun): object or thing that exists in reality
e.g. SUPPLIER, defined as an ENTITY that provides goods or services.



Learning Points

Modals for Ability, Can and Could

We use the modal verbs can and could to talk about skills or capacities.

When we refer to ability, we are talking about two different concepts

On the one hand, we have **general ability**, which refers to something that you learned how to do, and you do it whenever you want, like speaking French or swimming.

Examples:

He **can** set up a computer.

He **couldn't** set up the computer.

On the other hand, there is **specific ability**, which means something that you can or can't do in a particular situation. For instance, to be able to go somewhere or see something.

Examples:

She **can't** find the right language model.

She **couldn't** find the right language model.

Notice that normally we use **can** to talk about general ability in **the present**, and we use could to describe general ability in **the past**. However, for referring to **general ability in the future**, we use the expression *to be able to*.

Examples:

By December, you **will be able to** code in Java.

You **won't be able to** program in C+ in a weekend! It usually takes months.

Examples

- *I can see why you think that. People confuse those concepts easily.*
- *We can create data models to represent the business reality, providing a design for data storage location and its management.*
- *Now I see why I couldn't differentiate both concepts.*





Learning Points

Connectors for Expressing Result, for Reformulating and for Providing Alternatives

Connectors are words or expressions that link phrases or sentences together.

By using connectors, we can express relationships between ideas, making our writing easier to read, especially when referring to complex concepts.

Some connectors express the **result** or **consequence** of an action or situation. Observe:

We couldn't finish the presentation. We postponed the presentation.

*We couldn't finish the presentation, **therefore** we postponed it.*

*We couldn't finish the presentation, **as a result**, we postponed it.*

There are connectors that we can use for **reformulating** or further **explaining** a concept or complex idea.

For example:

*Debugging is the process of correcting program errors, **in other words**, it is how you repair what is wrong with your code.*

Finally, we use certain connectors when we want to **provide a different meaning**, contrast or **alternative** to the idea we are expressing.

For example:

*Certain automated systems can recognize some particular patterns, **on the other hand**, a normal human brain can effortlessly recognize millions of patterns.*

Examples

- *I didn't have to work, so I could study the whole weekend; as a result, I think I am ready.*
- *Well, I can't understand what ontologies and data models are. In other words, I think they are the same.*
- *On the other hand, we can create data models to represent the business reality, providing a design for data storage location and its management.*

